



Data Visualization and Prediction of DFU Healing: Wifl Component Outperforms the Composite Stage, and More Sophisticated Models Aren't Much Better



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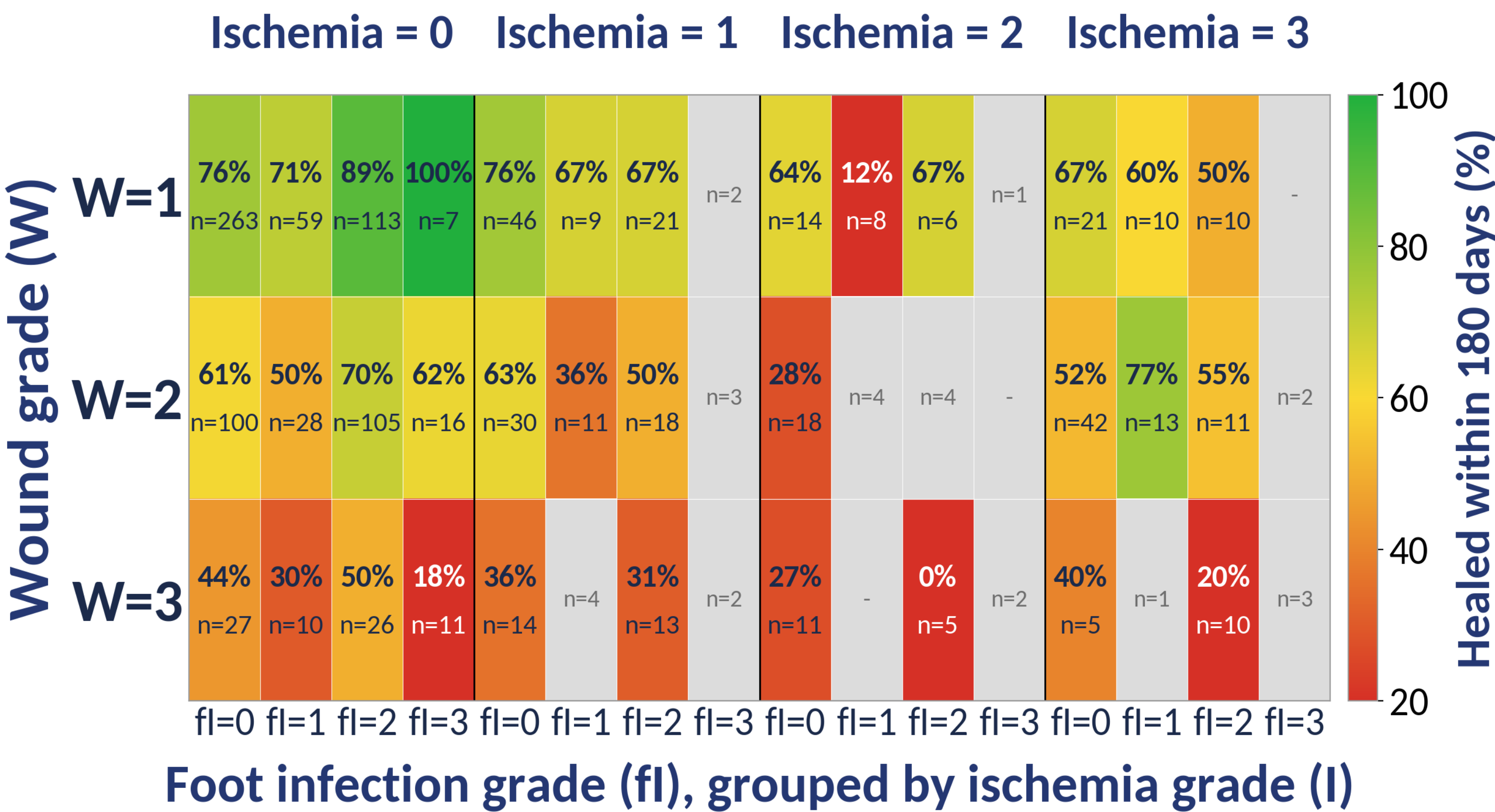
Background & Objectives

- Wifl staging was designed to predict limb threat at the population level, not whether an individual wound will heal.
- Objective 1: use data visualization to better understand patient progression and wound healing, allowing a clinician/patient to prognosticate.
- Objective 2: test whether any model, simple or complex, predicts healing better than the Wifl grades themselves.

Methods

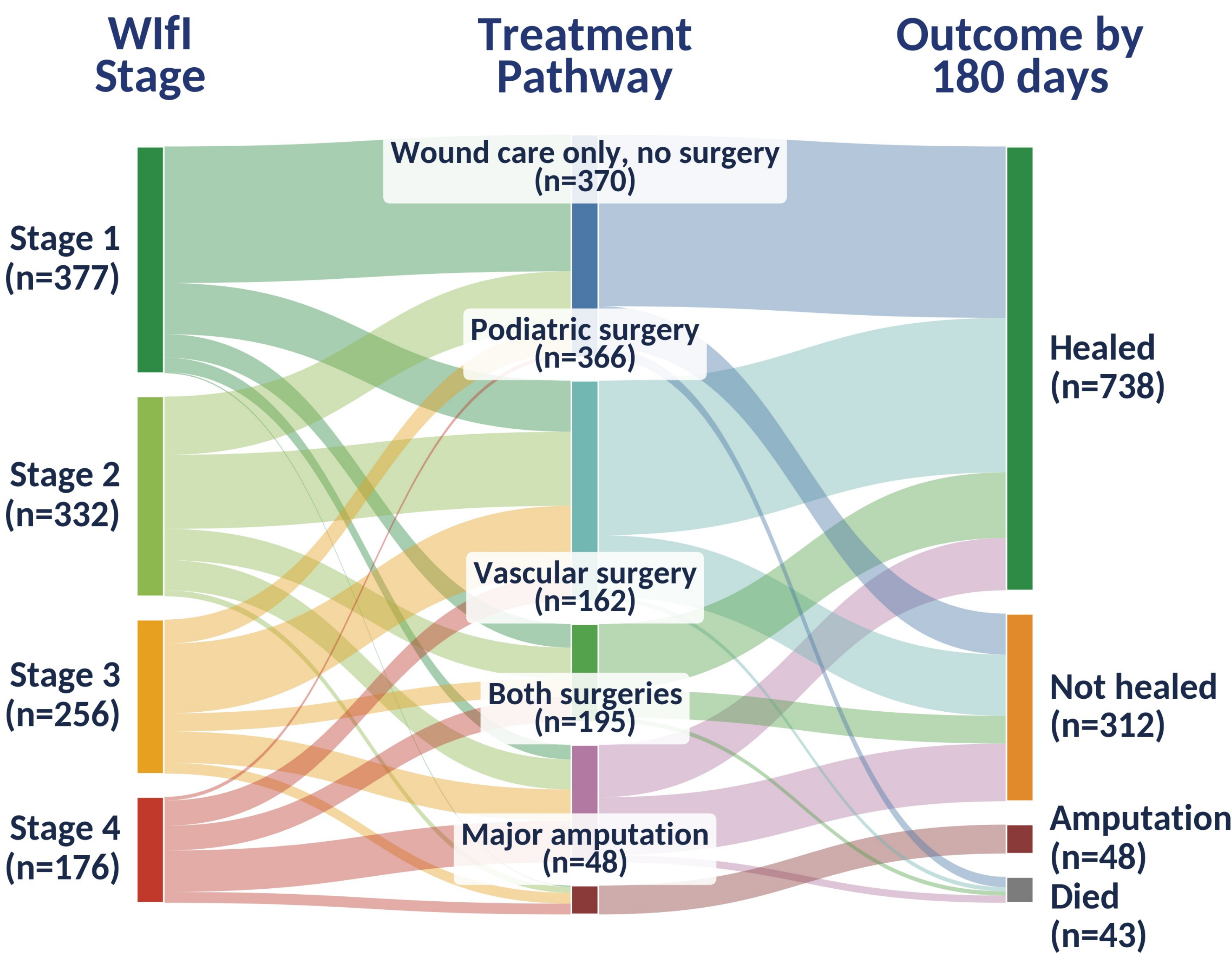
- JHH Diabetic Foot and Wound Center, 2012 to 2025.
- 1,255 DFUs (Wifl 1 to 4), 422 patients.
- 114 excluded for inadequate follow-up.
- Analysed: 1,141 wounds, 387 patients.
- Outcome: healed by 180 days. Split 755/386.

Visualizing Healing by Wifl



Actual healing rates within 180 days plotted on the canonical Wifl wound x ischemia x infection matrix (n=1,139). Cells with fewer than 5 wounds are shown in grey.

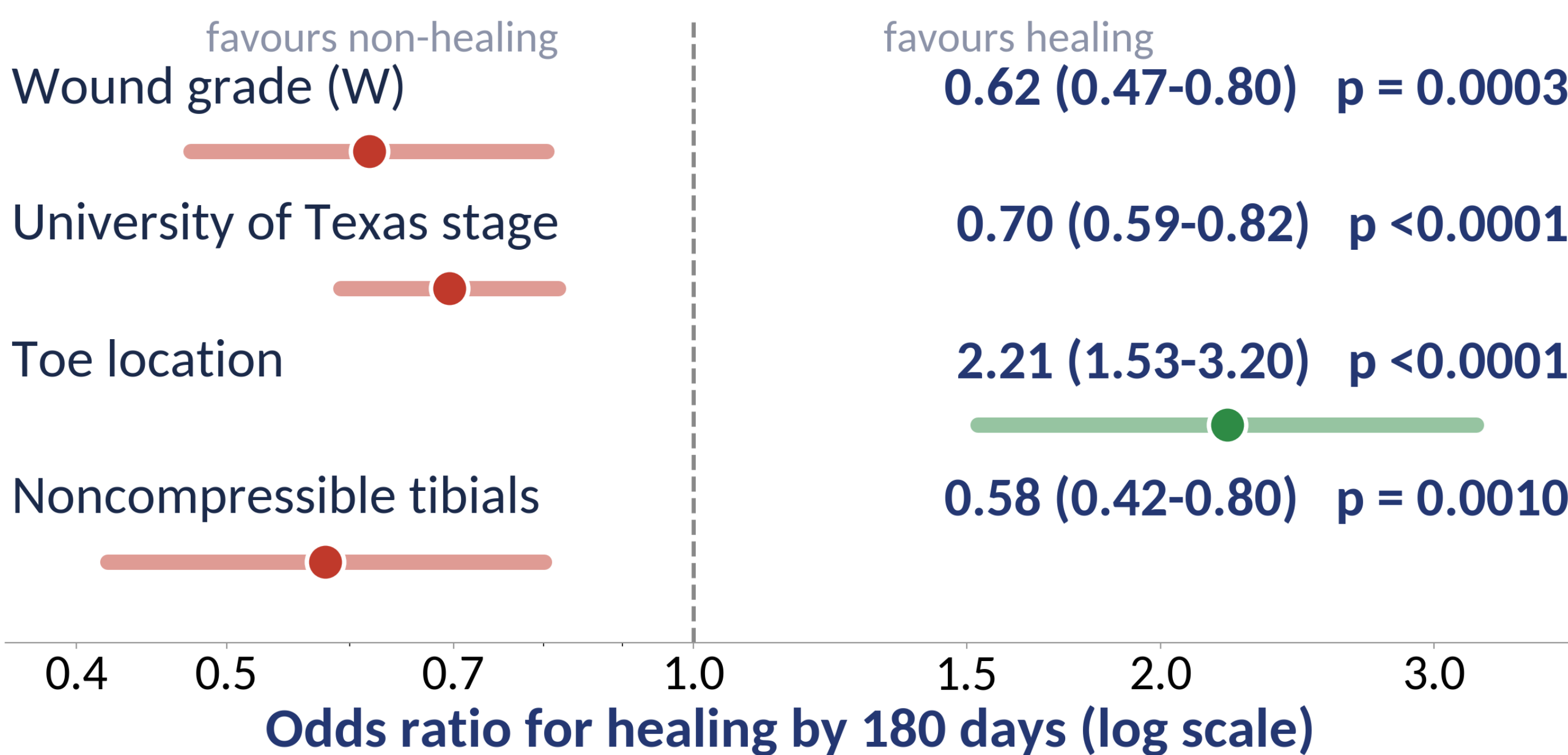
Visualizing Patient Progression



Patient treatment and outcome recorded jointly to 180 days (n=1,141). Died = death on or before the wound outcome date.

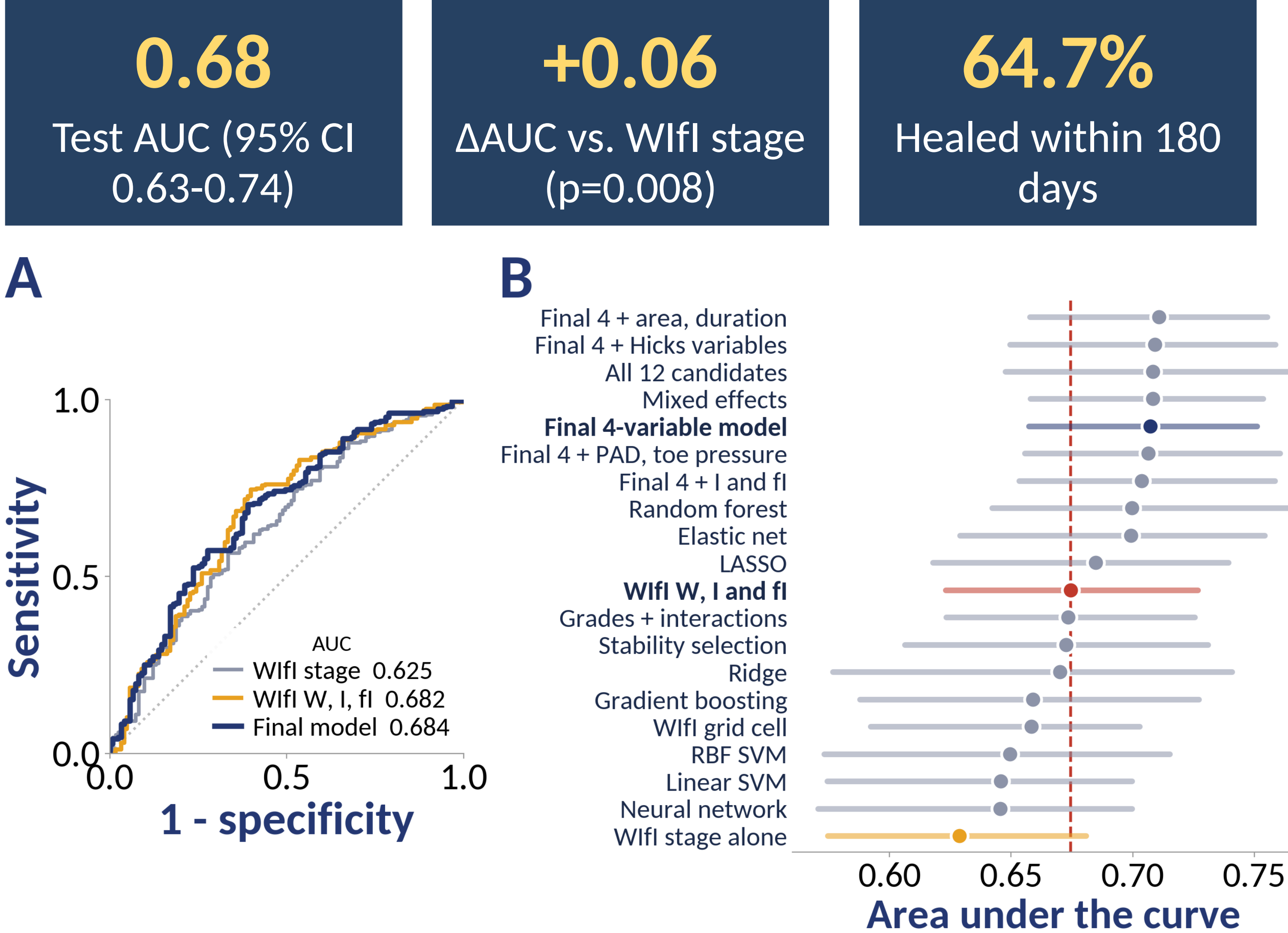
Key Wifl Model Findings

- Wifl stage 1 to 4: 75% to 46% healed ($p < 0.0001$).
- Ischemia I-0 to I-3: 70%, 58%, 41%, 54% ($p < 0.0001$).
- Noncompressible tibials: 57% vs. 71% ($p < 0.0001$).
- A final 4 variable model: AUC 0.684 vs. Wifl component stage 0.625, vs. Wifl composite 0.682.



Odds ratios on the native scale (per one grade or stage, or presence vs. absence) for the final four-variable model. The University of Texas stage grades infection and ischemia.

Model Performance



(A) ROC on the held-out test set (n=386). (B) Mean AUC over 60 repeated patient-level splits, all 20 approaches tested; dashed line marks the Wifl component grades.

		Observed by 180 days		Metric	Value
		Healed	Not healed		
Predicted	Healed	185 48% of test set	48 12% of test set	Sensitivity	70%
	Not healed	78 20% of test set	75 19% of test set	Specificity	61%
				PPV	79%
				NPV	49%
				Calibration slope	0.86
				Hosmer-Lemeshow p	0.004
				Brier score	0.199

Confusion matrix at 0.60.

Conclusions

- Wifl component grades (AUC 0.682) beat the composite stage (AUC 0.625). No more sophisticated model was significantly better (all $P > 0.05$), but a simple 4 variable model that trended better was created and provided.
- Better healing prediction needs a lot more data for sophisticated models to be useful.
- Data visualization may be useful prognostic tool until modeling catches up.



SCAN TO USE
THE CALCULATOR